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| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Session | Tree 1 | Question Information | Level 1 | Challenge 66 |
| <p><b>Question description</b></p> <p>A company has <math>n</math> employees, who form a tree hierarchy where each employee has a boss, except for the general director. Your task is to process <math>q</math> queries of the form: who is employee <math>x</math>'s boss <math>k</math> levels higher up in the hierarchy?</p> <p><b>Input</b></p> <p>The first input line has two integers <math>n</math> and <math>q</math>: the number of employees and queries. The employees are numbered <math>1, 2, \dots, n</math>, and employee <math>1</math> is the general director. The next line has <math>n-1</math> integers <math>e_2, e_3, \dots, e_n</math>: for each employee <math>2, 3, \dots, n</math> their boss. Finally, there are <math>q</math> lines describing the queries. Each line has two integers <math>x</math> and <math>k</math>: who is employee <math>x</math>'s boss <math>k</math> levels higher up?</p> <p><b>Output</b></p> <p>Print the answer for each query. If such a boss does not exist, print <math>-1</math>.</p> <p><b>Constraints</b></p> <ul style="list-style-type: none"><li>• <math>1 \leq n, q \leq 2 \cdot 10^5</math></li><li>• <math>1 \leq e_i \leq i-1</math></li><li>• <math>1 \leq x \leq n</math></li><li>• <math>1 \leq k \leq n</math></li></ul> |         |        |                      |         |              |
| <p>Logical Test Cases</p> <div><div><p>Test Case 1</p><p>INPUT (STDIN)</p><pre>5 3 1 3 2 2 4 2 1 2 1 3</pre><p>EXPECTED OUTPUT</p><pre>1 -1 -1</pre></div><div><p>Test Case 2</p><p>INPUT (STDIN)</p><pre>5 3 1 1 3 3 4 1 4 2 4 3</pre><p>EXPECTED OUTPUT</p><pre>3 1 -1</pre></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |        |                      |         |              |
| <p>Mandatory Test Cases</p> <div><p>Test Case 1</p><p>KEYWORD</p><pre>void link(int i,int j)</pre></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |        |                      |         |              |

Complexity Test Cases

|                       |
|-----------------------|
| Test Case 1           |
| CYCLOMATIC COMPLEXITY |
| 2                     |

|             |
|-------------|
| Test Case 2 |
| TOKEN COUNT |
| 420         |

|             |
|-------------|
| Test Case 3 |
| NLOC        |
| 50          |